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Tarea #1

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79/100%

Preguntas conceptuales

1. No. Ambas fuerzas la gravitacional y la eléctrica, son $\propto \frac{1}{d^2}$, entonces no se gana nada variando d .
Matemáticamente: ✓ 5/5

$$\frac{F_e}{F_g} = 2,3 \times 10^{39}$$

$$\Leftrightarrow \frac{\left(\frac{k_e q_1 q_2}{d^2} \right)}{\left(\frac{G m_1 m_2}{d^2} \right)} = 2,3 \times 10^{39}$$

$$\Leftrightarrow \frac{k_e q_1 q_2}{G m_1 m_2} = 2,3 \times 10^{39}$$

No hay dependencia de d .

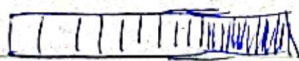
2. 1. La fuerza es de repulsión.
2. Por 3era Ley de Newton, las fuerzas experimentadas son igual en magnitud.

Entonces la opción c. ✓

3. La fuente del campo eléctrico → Más claridad
5/5

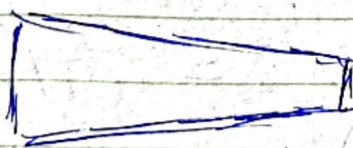
4. Ley de Gauss es útil para ~~si~~ calcular el campo eléctrico para distribuciones de carga simétricas
X
1/5

a.



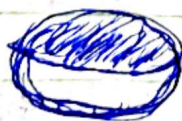
barra con densidad de carga $\sigma(x) \propto x$.

b.



barra con densidad de carga uniforme, pero ~~asimétrica~~ asimétrica.

c.

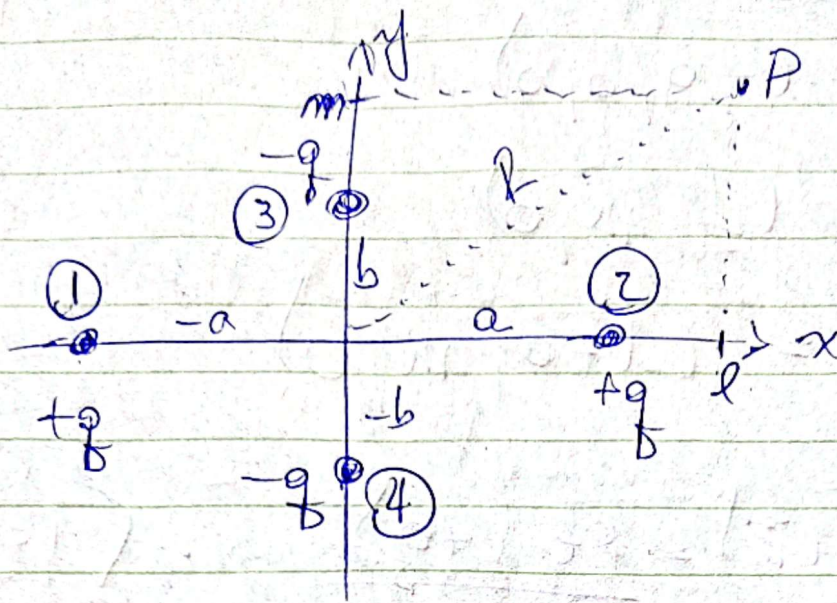


Media esfera

En todos estos casos, el $\vec{E}$ X no es uniforme para superficies gaussianas

Problema #1.

(a)



$$\vec{E}_i(\vec{r}) = k_e \frac{q_i}{|\vec{r} - \vec{r}_i|^3} (\vec{r} - \vec{r}_i)$$

$$\vec{E}(\vec{r}) = k_e \sum_{i=1}^4 \frac{q_i}{|\vec{r} - \vec{r}_i|^3} (\vec{r} - \vec{r}_i)$$

Diagram illustrating the vector relationship: $\vec{r}$ is the vector from origin O to point P, $\vec{r}_i$ is the vector from origin O to charge q_i , and $\vec{r} - \vec{r}_i$ is the vector from charge q_i to point P.

donde

$\vec{r}$: Vector posición de P.

$\vec{r}_i$: Vector posición de q_i

$\vec{r} - \vec{r}_i$: vector de $q_i \rightarrow P$.

Para ①.

$$\vec{r}_1 = (-a, 0, 0)$$

$$\vec{r} = (l, m, 0)$$

$$\vec{r} - \vec{r}_1 = (l+a, m, 0)$$

$$\vec{E}_1(\vec{r}) = \frac{k_e (+q)}{[(l+a)^2 + m^2]^{3/2}} \cdot \langle l+a, m, 0 \rangle$$

Para ②:

$$\vec{r}_2 = (a, 0, 0)$$

$$\vec{r} = (l, m, 0)$$

$$\vec{r} - \vec{r}_2 = (l-a, m, 0)$$

$$\vec{E}_2(\vec{r}) = \frac{k_e (+q)}{[(l-a)^2 + m^2]^{3/2}} \cdot \langle l-a, m, 0 \rangle$$

Para ③:

$$\vec{r}_3 = (0, b, 0)$$

$$\vec{r} = (l, m, 0)$$

$$\vec{r} - \vec{r}_3 = (l, m-b, 0)$$

$$\vec{E}_3(\vec{r}) = \frac{k_e (-q)}{[l^2 + (m-b)^2]^{3/2}} \cdot \langle l, m-b, 0 \rangle$$

Para ④:

$$\vec{r}_4 = (0, -b, 0)$$

$$\vec{r} = (l, m, 0)$$

$$\vec{r} - \vec{r}_4 = (l, m+b, 0)$$

$$\vec{E}_4(\vec{r}) = \frac{k_e (-q)}{[l^2 + (m+b)^2]^{3/2}} \cdot \langle l, m+b, 0 \rangle$$

$$\vec{E}(\vec{r})_x = \vec{E}_1(\vec{r})_x + \vec{E}_2(\vec{r})_x + \vec{E}_3(\vec{r})_x + \vec{E}_4(\vec{r})_x$$

Se pone cuando $E = E_1 + E_2 + E_3 + E_4$

$$E(\vec{r})_x = \frac{q}{4\pi\epsilon_0} \left[\frac{l+a}{[(l+a)^2 + m^2]^{3/2}} + \frac{l-a}{[(l-a)^2 + m^2]^{3/2}} - \frac{l}{[l^2 + (m-b)^2]^{3/2}} - \frac{l}{[l^2 + (m+b)^2]^{3/2}} \right]$$

$$\vec{E}(\vec{r})_y = \vec{E}_1(\vec{r})_y + \vec{E}_2(\vec{r})_y + \vec{E}_3(\vec{r})_y + \vec{E}_4(\vec{r})_y$$

$$\vec{E}(\vec{r})_y = \frac{q}{4\pi\epsilon_0} \left[\frac{m}{[(l+a)^2 + m^2]^{3/2}} + \frac{m}{[(l-a)^2 + m^2]^{3/2}} \right]$$

$$= \frac{(m-b)}{[l^2 + (m-b)^2]^{3/2}} - \frac{(m+b)}{[l^2 + (m+b)^2]^{3/2}} \quad \checkmark$$

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$$\vec{E}(\vec{r})_z = 0 \quad \checkmark$$

(b)

Está tudo $\checkmark$ x6

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(C) para $m = 0$

$$E(r)_x = Kq \left(\frac{l+a}{(l^2+a^2)^{3/2}} + \frac{l-a}{[(l-a)^2]^{3/2}} - \frac{l}{(l^2+b^2)^{3/2}} - \frac{l}{(l^2+b^2)^{3/2}} \right)$$

$$E(r)_x = Kq \left(\underbrace{\frac{1}{(l+a)^2}}_{(*)} + \underbrace{\frac{1}{(l-a)^2}}_{(**)} - \underbrace{\frac{2l}{(l^2+b^2)^{3/2}}}_{\cancel{***}} \right)$$

$$R = \sqrt{l^2 + m^2} \rightarrow R = \sqrt{l^2}$$

$$\rightarrow R = l$$

$$(*) \frac{1}{(l+a)^2} = \frac{1}{(l(1+\frac{a}{l}))^2} = l^{-2} \left(1 + \frac{a}{l}\right)^{-2}$$

$$l^{-2} \left(1 + \frac{a}{l}\right)^{-2} = l^{-2} \left(1 - \frac{2a}{l} + \frac{3a^2}{l^2} \dots\right)$$

$$= \frac{1}{l^2} - \frac{2a}{l^3} + \frac{3a^2}{l^4} \dots$$

$$(**) \frac{1}{(l-a)^2} = l^{-2} \left(1 - \frac{a}{l}\right)^{-2}$$

$$= l^{-2} \left(1 + \frac{2a}{l} + \frac{3a^2}{l^2} \dots\right)$$

$$= \left(\frac{1}{l^2} + \frac{2a}{l^3} + \frac{3a^2}{l^4} \right)$$

$$\textcircled{\otimes} : \frac{l}{(l^2 + b^2)^{3/2}} = \frac{l}{l^3 \left(1 + \frac{b^2}{l^2}\right)^{3/2}} = \frac{1}{l^2} \left(1 + \frac{b^2}{l^2}\right)^{-3/2}$$

$$= \frac{1}{e^2} \left(1 - \frac{3}{2} \frac{b^2}{e^2} + \dots \right)$$

$$= \frac{1}{l^2} - \frac{3}{2} \frac{b^2}{l^4} + \dots$$

Entrances $\otimes + \otimes + -2\otimes\otimes$

$$\frac{1}{l^2} - \frac{2a}{l^3} + \frac{3a^2}{l^4}$$

$$= \frac{1}{e^2} + \frac{2a}{e^3} + \frac{3a^2}{e^4}$$

$$-\frac{2}{t^2} + 3 \frac{t^2}{t^4}$$

$$0 + 0 + \frac{3a^2 + 3a^2 + 3b^2}{l^4} = \frac{3(a^2 + b^2)}{l^4}$$

$$\vec{E}(m=0)_x = Kq \frac{3(2a^2 + b^2)}{l^4} \quad \text{Lo mismo para } E(m=0)_y.$$

Problema #2

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(a)

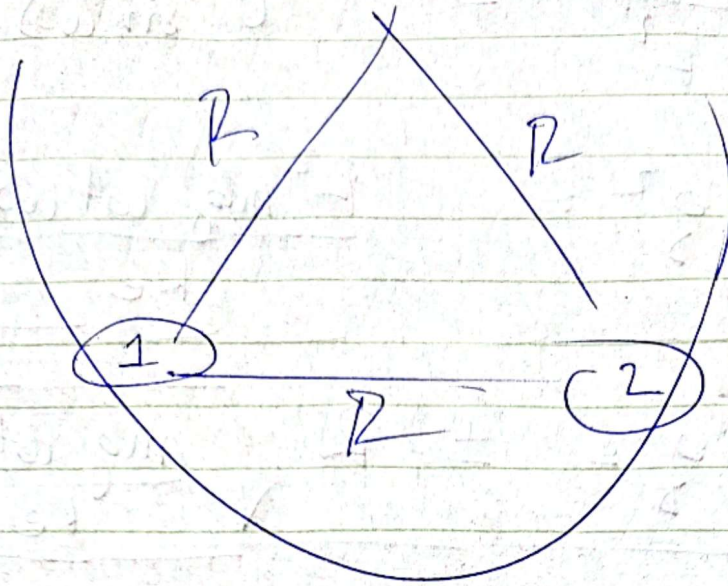
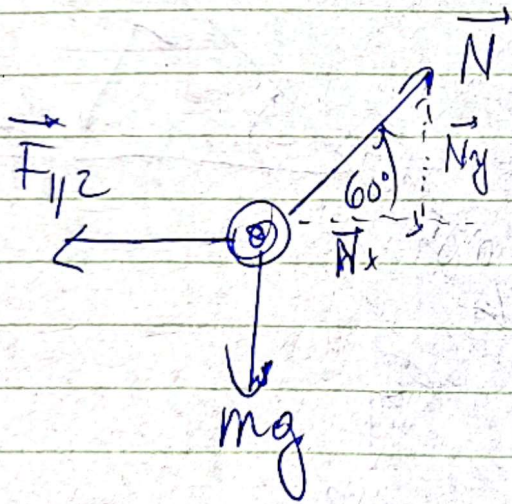


Diagrama de cuerpo libre para ①:



$$\left\{ \begin{array}{l} \sum \vec{F}_x: \vec{N}_x - \vec{F}_{12} = 0 \quad (1) \\ \sum \vec{F}_y: \vec{N}_y - m\vec{g} = 0 \quad (2) \end{array} \right.$$

De ②:

$$N_y = mg$$

$$\Rightarrow N \sin(60) = mg$$

$$N = \frac{mg}{\sin(60)} \quad (3)$$

De ①:

$$\vec{F}_{12} = \vec{N}_x$$

$$\frac{K e q^2}{r^2} = \cos(60) N \quad (4)$$

(3) en (4):

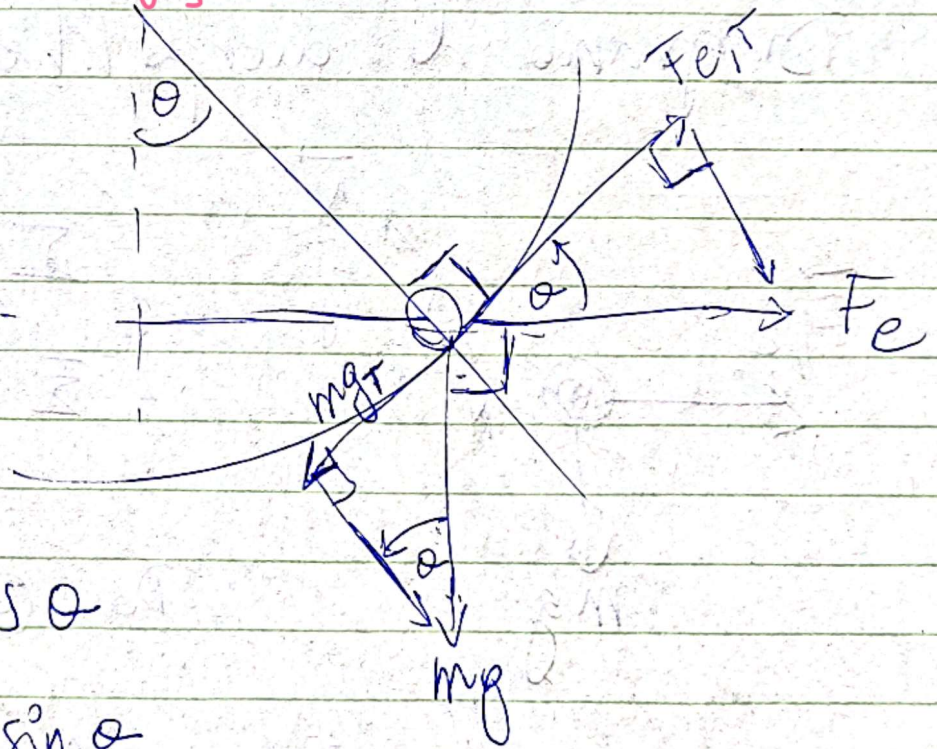
$$\frac{k_e q^2}{R^2} = mg \frac{\cos(60)}{\sin(60)}$$

$$\Rightarrow q^2 = \frac{R^2 mg \cot(60)}{k_e}$$

$$\Rightarrow \left[q = \pm R \sqrt{\frac{mg \cot(60)}{k_e}} \right]$$

$$q = R \sqrt{\frac{4\pi\epsilon_0 mg}{\sqrt{3}}}$$

(b)



$$T \cos \theta = Fe \cos \theta$$

$$mg \sin \theta = mg \sin \theta$$

$$\Sigma F_T = m a_T$$

$$Fe \cos \theta - mg \sin \theta = m R \ddot{\theta}$$

y ya de aquí está rudo seguir...

||

Problema # 3.

De Gauss:

$$A_{\text{surf}} = 4\pi r^2$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\Rightarrow E (4\pi r^2) = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\Rightarrow E(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q_{\text{enc}}}{r^2}$$

$$\rho = \frac{Q_{\text{enc}}}{V} \rightarrow Q_{\text{enc}} = \rho V$$

$$dQ_{\text{enc}}(r') = \rho dV$$

$$Q_{\text{enc}}(r') = \int_V \rho dV$$

$$Q_{\text{enc}}(r') = \int_0^{\pi} \int_0^{2\pi} \int_0^{r'} \frac{\rho}{\pi a^3} e^{-\frac{2r}{a}} r'^2 \sin\theta dr' d\phi d\theta$$

$$\int_0^{\pi} \int_0^{2\pi} \sin\theta d\phi d\theta$$

$$Q_{\text{enc}}(r') = \int_0^{r'} \frac{\rho}{\pi a^3} e^{-\frac{2r}{a}} r'^2 dr' \cdot 4\pi$$

$$\rho_{enc}(r') = \frac{4\pi q}{\pi a^3} \int_0^{r'} e^{-2r'/a} r'^2 dr'$$

I_1

Ahora hay que integrar por partes.

$$\begin{array}{ll} u = r'^2 & v = -\frac{a}{2} e^{-2r'/a} \\ \downarrow & \uparrow \\ du = 2r' dr' & dv = e^{-2r'/a} dr' \end{array}$$

$$I_1 = uv - \int v du$$

$$I_1 = r'^2 \cdot \frac{(-a)}{2} e^{-2r'/a} + \int ar' e^{-2r'/a} dr'$$

I_2

I_2 también va por partes.

$$\begin{array}{ll} u = r' & v = -\frac{a}{2} e^{-2r'/a} \\ \downarrow & \uparrow \\ du = dr' & dv = e^{-2r'/a} dr' \end{array}$$

$$I_2 = ur - \int r du$$

$$I_2 = -\frac{a}{2} r' e^{-2r'/a} - \int_0^{r'} -\frac{a}{2} e^{-2r'/a} dr'$$

$$I_2 = -\frac{a}{2} r' e^{-2r'/a} + \frac{a}{2} \int_0^{r'} e^{-2r'/a} dr$$

$$I_2 = -\frac{a}{2} r' e^{-2r'/a} + \frac{a}{2} \cdot -\frac{a}{2} e^{-2r'/a}$$

$$I_2 = \left[-\frac{a}{2} r' e^{-2r'/a} - \frac{a^2}{4} e^{-2r'/a} \right]_0^r$$

$$I_2 = -\frac{a}{2} r e^{-2r/a} - \frac{a^2}{4} e^{-2r/a} + \frac{a^2}{4}$$

Entonces $I_1 = -\frac{a}{2} r^2 e^{-2r/a} + a \cdot I_2$

$$I_1 = -\frac{a}{2} r^2 e^{-2r/a} - \frac{a}{2} r e^{-2r/a} - \frac{a^2}{4} e^{-2r/a} + \frac{a^3}{4}$$

$$I_1 = \frac{a}{2} \left(-r^2 e^{-2r/a} - a r e^{-2r/a} - \frac{a^2}{2} e^{-2r/a} + \frac{a^2}{2} \right)$$

$$Q_{enc}(a) = \frac{4q}{a^3} \cdot I_1$$

$$Q_{enc}(r) = \frac{2q}{a^2} \left(-r^2 e^{-2r/a} - a r e^{-2r/a} - \frac{a^2}{2} e^{-2r/a} + \frac{a^2}{2} \right)$$

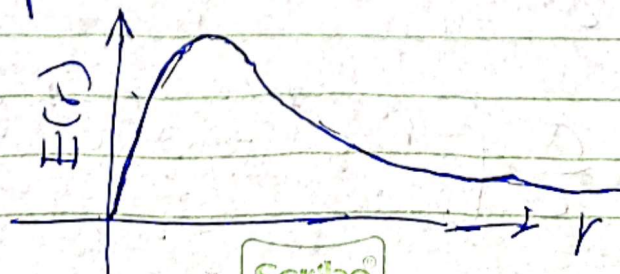
$$Q_{enc}(r) = q \left(\frac{-2r^2}{a^2} e^{-2r/a} - \frac{2r}{a} e^{-2r/a} - e^{-2r/a} + 1 \right)$$

$$Q_{enc}(r) = q \left(1 - e^{-2r/a} \left(1 + \frac{2r}{a} + \frac{2r^2}{a^2} \right) \right)$$

$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0} \frac{Q_{enc}}{r^2}$$

$$E(r) = \frac{q}{4\pi\epsilon_0 r^2} \left(1 - e^{-2r/a} \left(1 + \frac{2r}{a} + \frac{2r^2}{a^2} \right) \right)$$

(b) El gráfico de $\vec{E}(r)$ se ve así:



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Problema #4

$$A_{\text{est}} = 4\pi r^2$$

De Gauss:

$$V_{\text{est}} = \frac{4\pi r^3}{3}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

Para una esfera gaussiana:

$$E \cdot (4\pi r^2) = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\Leftrightarrow E = \frac{1}{4\pi\epsilon_0 r^2} \cdot Q_{\text{enc}}$$

Zona 1: $0 < r' < R_1$

$$\vec{E}_1 = \vec{0} \quad \checkmark +1$$

Zona 2: $R_1 < r' < R_2$

$$\rho_2 = \frac{Q_2}{V_2} = \frac{-3q}{\frac{4}{3}\pi(R_2^3 - R_1^3)} \quad \checkmark$$

$$\rho_2 = \frac{Q_2'}{V_2'} \rightarrow Q_2' = \rho_2 V_2'$$

↓

$$Q_2' = \frac{Q}{V_2} \cdot V_2'$$

$$Q_2' = \frac{-3q}{\frac{4}{3}\pi(R_2^3 - R_1^3)} \cdot \frac{4}{3}\pi(r^3 - R_1^3)$$

$$Q_2' = -3q \cdot \frac{(r^3 - R_1^3)}{(R_2^3 - R_1^3)}$$

Ahora hay que sumar la carga de zona 1.

$$Q_{enc}^{(2)} = Q_{z1} + Q_2'$$

$$Q_{enc}^{(2)} = q - 3q \frac{(r^3 - R_1^3)}{(R_2^3 - R_1^3)}$$

Entonces

$$E_2(r) = \frac{1}{4\pi\epsilon_0 r^2} \cdot \left(q - 3q \frac{(r^3 - R_1^3)}{(R_2^3 - R_1^3)} \right) \checkmark$$

+5

Zona 3:

$$R_2 < r' < R_3$$

$$\varphi_{enc}^{(3)} = q - 3q = -2q$$

$$E_3(r') = 0 \cdot \frac{1}{4\pi\epsilon_0 r'^2} \cdot (-2q) \quad \times \quad 0$$

Zona 4:

$$R_3 < r' < R_4$$

$$\rho_4 = \frac{\varphi_4}{V_4} = \frac{2q}{\frac{4}{3}\pi (R_4^3 - R_3^3)}$$

$$\rho_4 = \frac{\varphi_4'}{V_4'} \rightarrow \varphi_4' = \rho_4 V_4'$$

↓

$$\varphi_4' = \frac{\varphi_4 \cdot V_4'}{V_4}$$

$$\varphi_4' = \frac{2q}{\cancel{\frac{4}{3}\pi (R_4^3 - R_3^3)}} \cdot \cancel{\frac{4}{3}\pi (r'^3 - R_3^3)}$$

$$\varphi_4' = 2q \cdot \frac{(r'^3 - R_3^3)}{(R_4^3 - R_3^3)}$$

Ahora hay que sumarle las cargas de zona 1, zona 2 y zona 3.

$$(4) \quad Q_{enc} = q + (-3q) + (-4q) + Q_4'$$

$$(4) \quad Q_{enc} = -6q + 2q \frac{(r_1^3 - r_3^3)}{(r_4^3 - r_3^3)}$$

Entonces

$$E_4(r) = \frac{1}{4\pi\epsilon_0 r^2} \cdot \left(-6q + 2q \frac{(r_1^3 - r_3^3)}{(r_4^3 - r_3^3)} \right)$$

+5

Zona externa: $r' > r_4$

$$Q_{enc} = Q_{z1} + Q_{z2} + Q_{z3} + Q_{z4}$$

$$Q_{enc} = q - 3q - 4q + 2q = -4q$$

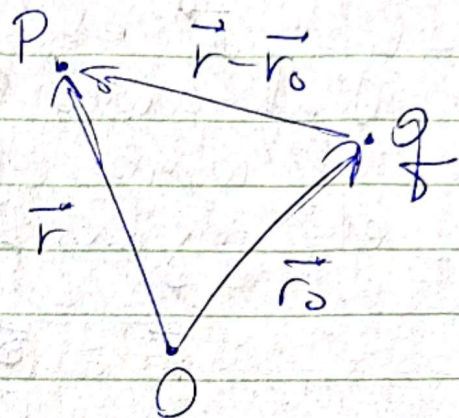
$$E(r') = \frac{1}{4\pi\epsilon_0 r'^2} \cdot (-4q)$$

$$= \frac{-q}{\pi\epsilon_0 r'^2}$$

+3

Problema # 5.

(a) Carga puntual:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{|\vec{r} - \vec{r}_0|^3} (\vec{r} - \vec{r}_0)$$


Varias cargas

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i}{|\vec{r} - \vec{r}_i|^3} (\vec{r} - \vec{r}_i)$$

Para una distribución volumétrica

$$d\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{dq'}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho dV'}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$$

$$\rho = \frac{Q}{V}$$

$$\rho = \frac{dq'}{dV'}$$

$$\rightarrow dq' = \rho dV'$$

Donde el prima (') indica elemento diferencial

En esa expresión

$$\rho = \rho(\vec{r}).$$

y $\vec{r}$ es el radio vector del origen al punto P. (donde queremos determinar el campo Eléctrico). Pero P es, justamente el origen. Entonces

$$\vec{r} = \vec{0}.$$

Por lo que:

$$\vec{E}(\vec{0}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}') dV'}{|\vec{r}'|^3} (-\vec{r}')$$

Usando esféricas:

$$x = r \sin\theta \cos\phi$$

$$r: 0 \leq r$$

$$y = r \sin\theta \sin\phi$$

$$\theta: [0, \pi]$$

$$z = r \cos\theta$$

$$\phi: [0, 2\pi]$$

$$dV = r^2 \sin\theta dr d\theta d\phi$$

Forces

$$\vec{E}(0) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_0 \frac{r}{R^2} (\vec{r})}{r^3} \cdot r^2 \sin\theta \, dr \, d\theta \, d\phi$$

Con $r: R_1 \rightarrow R_2$

$\theta: \pi/2 \rightarrow \pi$

$\phi: 3\pi/2 \rightarrow 2\pi$

$$\vec{E}(0) = \frac{-\rho_0}{4\pi\epsilon_0 R^2} \int_{R_1}^{R_2} \int_{\pi/2}^{\pi} \int_{3\pi/2}^{2\pi} \vec{r} \sin\theta \, d\phi \, d\theta \, dr$$

$$\vec{r} = (x, y, z)$$

$$\vec{r} = r (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$$

$$E(0)_x = \frac{-\rho_0}{4\pi\epsilon_0 R^2} \int_{R_1}^{R_2} r \, dr \int_{\pi/2}^{\pi} \sin^2\theta \, d\theta \int_{3\pi/2}^{2\pi} \cos\phi \, d\phi$$

$$E(0)_y = E(0)_x, \text{ por simetria.}$$

$$E(0)_z = \frac{-\rho_0}{4\pi\epsilon_0 R^2} \int_{R_1}^{R_2} r \, dr \int_{\pi/2}^{\pi} \sin\theta \cos\theta \, d\theta \int_{3\pi/2}^{2\pi} d\phi$$

$$E(\omega)_x = \frac{-\rho_0}{4\pi\epsilon_0 R_2^2} \left(\frac{R_2^2 - R_1^2}{2} \right) \left(\frac{\pi}{4} \right) (1)$$

$$E(\omega)_x = \frac{-\rho_0 (R_2^2 - R_1^2)}{8\pi\epsilon_0 R_2^2} \cdot \frac{\pi}{4} = E(\omega)_y$$

$$E(\omega)_z = \frac{-\rho_0}{4\pi\epsilon_0 R_2^2} \left(\frac{R_2^2 - R_1^2}{2} \right) \left(-\frac{1}{2} \right) \left(\frac{\pi}{2} \right)$$

$$E(\omega)_z = \frac{+\rho_0 (R_2^2 - R_1^2)}{8\pi\epsilon_0 R_2^2} \cdot \frac{\pi}{4}$$

Forces

$$\vec{E}(\omega) = \frac{\rho_0 (R_2^2 - R_1^2)}{8\pi\epsilon_0 R_2^2} \frac{1}{4} (-1, 1, 1)$$

$$\boxed{\vec{E}(\omega) = \frac{\rho_0 (R_2^2 - R_1^2)}{32\epsilon_0 R_2^2} (-1, 1, 1)}$$

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$$(b) \quad \vec{E} = \frac{\vec{F}}{q}$$

$$\vec{F} = q \vec{E}$$

$$\vec{F} = \frac{-3q \rho_0 (R_2^2 - R_1^2)}{32 \epsilon_0 R_2^2} (-1, 1, 1)$$

$$\vec{F} = \frac{q \rho_0 (R_2^2 - R_1^2)}{32 \epsilon_0 R_2^2} \checkmark (1, -1, -1)$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $\quad \quad \quad j \quad \quad k$